

Supporting information:

Optical Printing of Single Au Nanostars

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PVP-coated AuNS Synthesis

Materials

Tetrachloroauric (III) acid (HAuCl_4), polyvinylpyrrolidone with an average molecular weight of 10,000 (PVP) and trisodium citrate tribasic dihydrate (Na_3Cit) were purchased from Sigma-Aldrich. N,N-dimethylformamide (DMF) was supplied by Biopack, and absolute ethanol by Cicarelli. All the chemical reagents were employed as received. Deionized water (18 M Ω .cm) was used for all preparations.

Synthesis of PVP-coated seeds

PVP-coated seeds were synthesized following a previously seeded growth reported method, stopping the growth cycles when obtaining a suspension of ~ 36 nm diameter spherical nanoparticles coated with citrate.³⁵ To prepare the PVP-coated seeds, 2 mL of a PVP 8.5 mM was added dropwise to 15 mL of this seeds suspension and left stirring overnight. Then, it was centrifuged at 4500 rpm for 90 min, the supernatant was removed, and the nanoparticles were redispersed in 15 mL of ethanol.

Synthesis of PVP-coated AuNS

AuNS were synthesized following a previously reported procedure.³⁶ Briefly, 109 μL of 50 mM HAuCl_4 aqueous solution was added to 20 mL of a 10 mM PVP solution in DMF. Once the Au(III) charge-transfer-to-solvent absorption band at 325 nm disappeared, 304, 152 or 76 μL of PVP-coated seeds was added under incessant stirring and left reacting overnight. The three different amounts of seeds correspond to three different ratios between the concentration of gold precursor and seeds ($R = 5$, $R = 10$ and $R = 20$, respectively), leading to the different sized AuNS (85, 115 and 130 nm respectively). The product was centrifuged

at 4500 rpm for 60 min, the supernatant was removed, and the particles were redispersed in ethanol. The washing procedure was repeated four times, the suspension being redispersed for the last time in water.

Substrate functionalization for Optical Printing and electrostatic attachment of AuNS

Glass substrates were functionalized using a layer-by-layer deposition of polyelectrolytes. Before functionalization, the glass substrates (Paul Marienfeld) were cleaned through the following steps: sonication for 10 min in Hellmanex (0.2% v/v), thorough rinsing with Milli-Q water, rinsing with acetone, drying at 80 °C, and plasma cleaning for 3 min. Then, the substrate was positively functionalized by immersion for 15 min in a 1 mg/mL in 0.5 M NaCl solution of poly(diallyldimethylammonium chloride) (PDPA, Sigma-Aldrich, $M_w = 400.000\text{--}500.000$). After that, the glasses were rinsed with Milli-Q water several times. To obtain a negatively charged surface, an additional immersion for 15 min in a 1 mg/mL in 0.5 M NaCl solution of sodium polystyrene sulfonate (PSS, Sigma Aldrich, $M_w = 70.000$) was performed, followed by a final copious rinsing in Milli-Q water.

Characterization of single AuNS

Dark-field microscopy and spectroscopy of single AuNS deposited on glass were performed with an unpolarized infrared lamp and unpolarized detection. Dark-field images were acquired using a CMOS color digital camera (Canon EOS-500D). Scattering spectra were obtained in a spectrometer with 150 lines/mm grating (Shamrock 500i, Andor) and EMCCD camera (Andor Ixon EM+ 885). Scanning electron microscopy images of the optically printed AuNS were obtained in a Carl Zeiss Crossbeam 350 SEM-FIB at Centro Atómico

Bariloche, CNEA. The electrostatically attached AuNS were characterized in a Carl Zeiss LEO FE-SEM microscope, at the CMA, Facultad de Ciencias Exactas y Naturales, UBA.

Experimental Set up

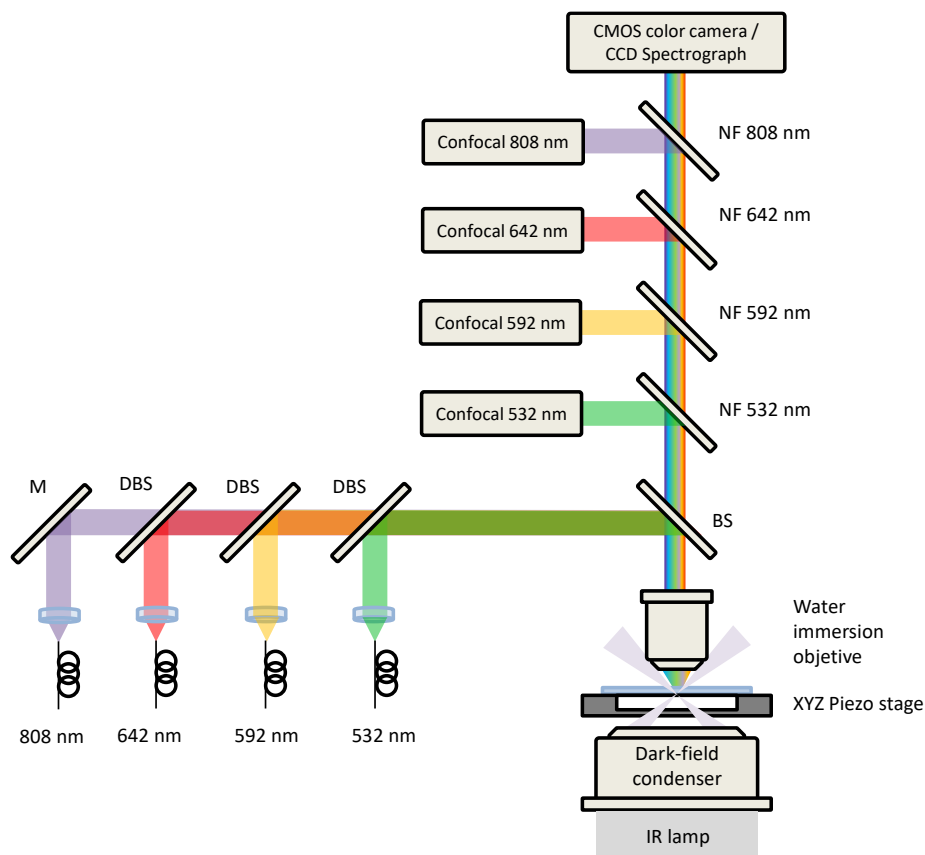


Figure S1. Schematic representation of the optical microscope with four laser wavelengths.

The microscope has four CW lasers: 532 nm (Laserquantum Ventus) 592 and 640 nm (MPB) and 808 nm (Thorlabs +Lumics), photodiodes and an XYZ piezoelectric stage (PI P-545) to perform the automated optical printing. The set-up is described on Figure S1.

Irradiance characterization:

The irradiance profile is considered Gaussian for the four lasers used. The beam waist ω of each laser is informed in Table S1. It was measured using confocal images of single NPs.

Then, the irradiance at the sample position (I) is computed by $I = \frac{2 \cdot P \cdot T}{\pi \omega^2}$, where P is the laser power measured at the back pupil of the objective and T is the transmission of the objective at the laser wavelength. Table S1 shows all the parameters. Printing P and printing I are the experimental parameters used during the Optical Printing experiments.

Laser wavelength (nm)	T	ω (nm)	Printing P (mW)	Printing I (mW. μm^{-2})
532	0.47 ± 0.01	268 ± 10	3.2 ± 0.2	13.3 ± 1.3
592	0.47 ± 0.01	310 ± 10	1.7 ± 0.2	5.3 ± 0.7
642	0.47 ± 0.01	312 ± 10	1.6 ± 0.2	4.9 ± 0.7
808	0.25 ± 0.01	381 ± 10	2.6 ± 0.2	2.8 ± 0.3

Table S1. Beam parameters corresponding to the Optical Printing experiments.

Characterization of electrostatically attached AuNSs

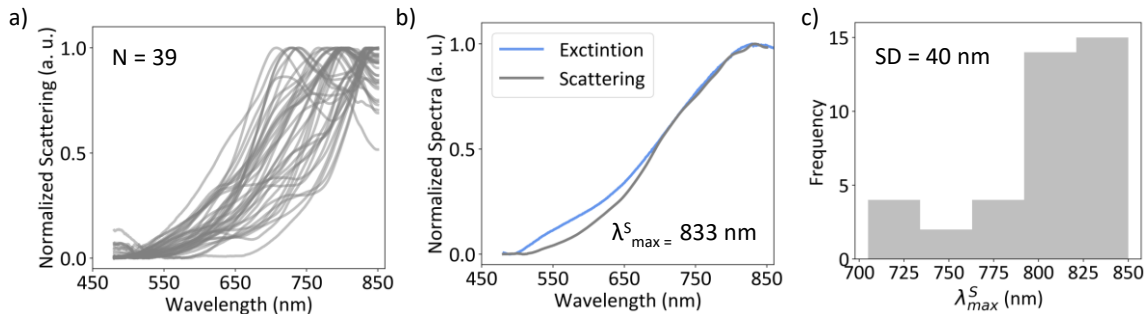


Figure S2. a) Normalized Scattering of 39 single AuNS immersed on water and deposited on glass by drop-casting. b) Normalized extinction spectra of the colloid (blue curve) and

average scattering spectra of 39 single AuNS (grey curve). The maximum average scattering ($\lambda_{\max}^S$) was 833 nm. c) Histogram of $\lambda_{\max}^S$ of the 39 single NS scattering maximum. The standard deviation was 40 nm.

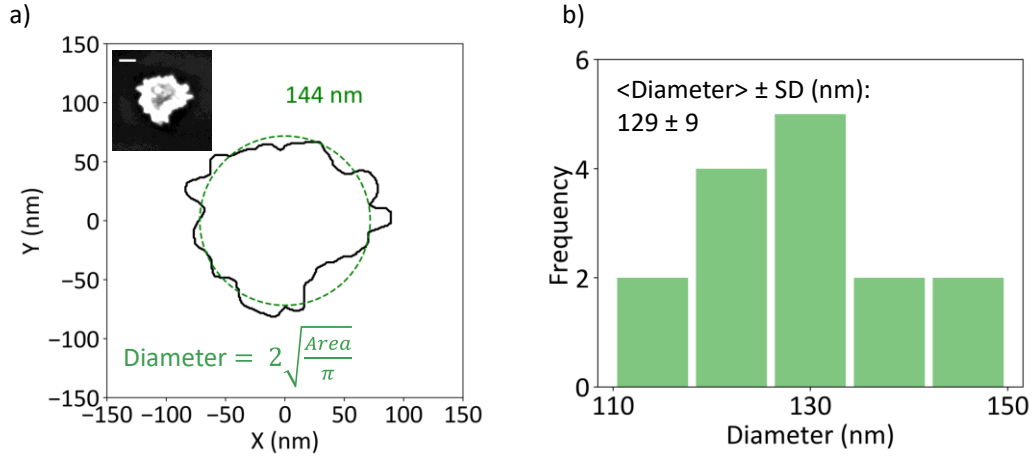


Figure S3. a) Measurement of AuNS diameter from the area of a SEM image. Scale bar of SEM image on inset: 50 nm. b) Histogram of AuNS diameters.

Classification of the outcomes after optical printing in three groups

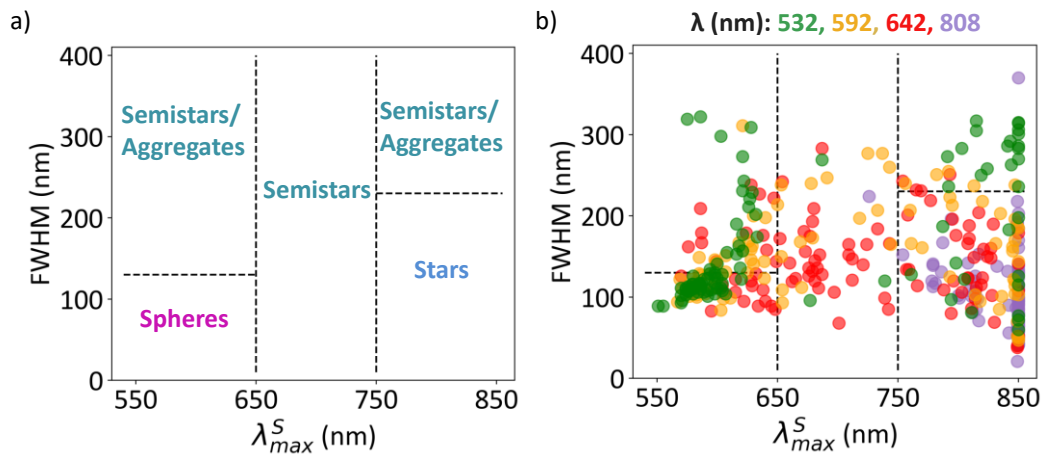


Figure S4. a) Regions delimited by the FWHM (nm) and $\lambda_{\max}^S$ for the classification of the NPs outcomes by optical printing. b) 100 outcomes at each laser wavelength.

Reshaping of AuNSs by irradiation

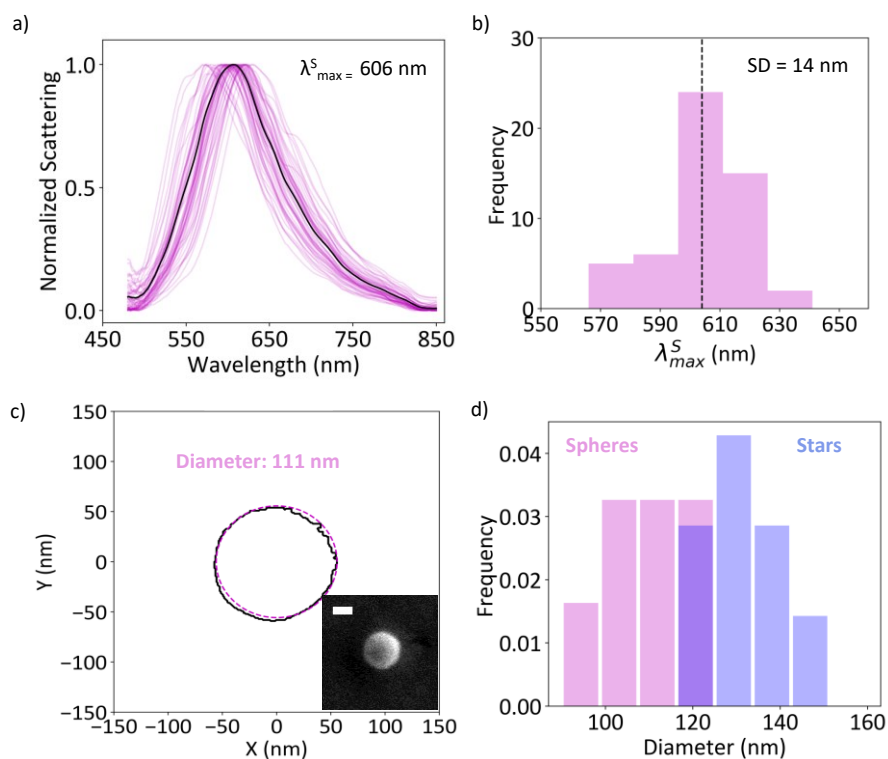


Figure S5. a) Scattering spectra of reshaped AuNS to spheres. The average scattering (black curve) presents a maximum wavelength ($\lambda_{\max}^S$) on 606 nm. b) Histogram of $\lambda_{\max}^S$. The standard deviation was 14 nm. c) Measurement of the reshaped NPs diameter. Scale bar of SEM image on inset: 50 nm. d) Histogram of diameters of the original AuNS (blue) and the reshaped NPs to spheres (magenta).

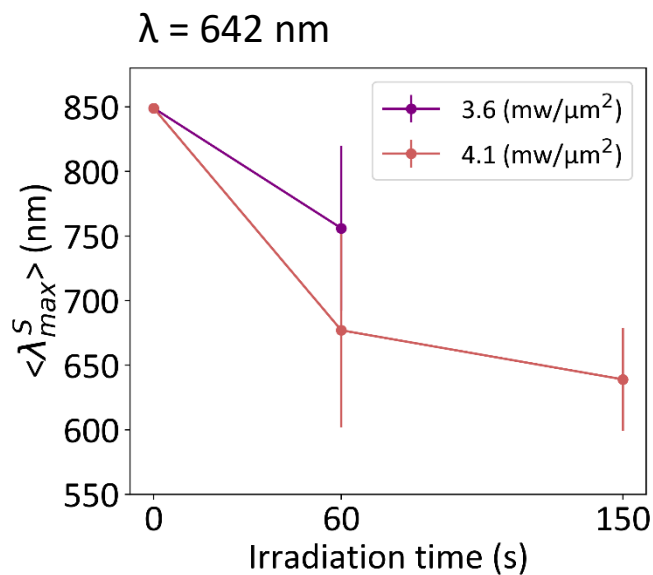


Figure S6. Reshaping of AuNSs at 642 nm. Final maximum wavelength resonance vs. irradiation time at two irradiances. Each dot is the average of the scattering maximum wavelength from nine individuals reshaped AuNSs. It could be seen that at higher irradiance and higher irradiation time, the scattering presents more blueshift respect to the original AuNS. Note that in this case the reshaping was not enough to transform each AuNS into a sphere.

Temperature rise by irradiation

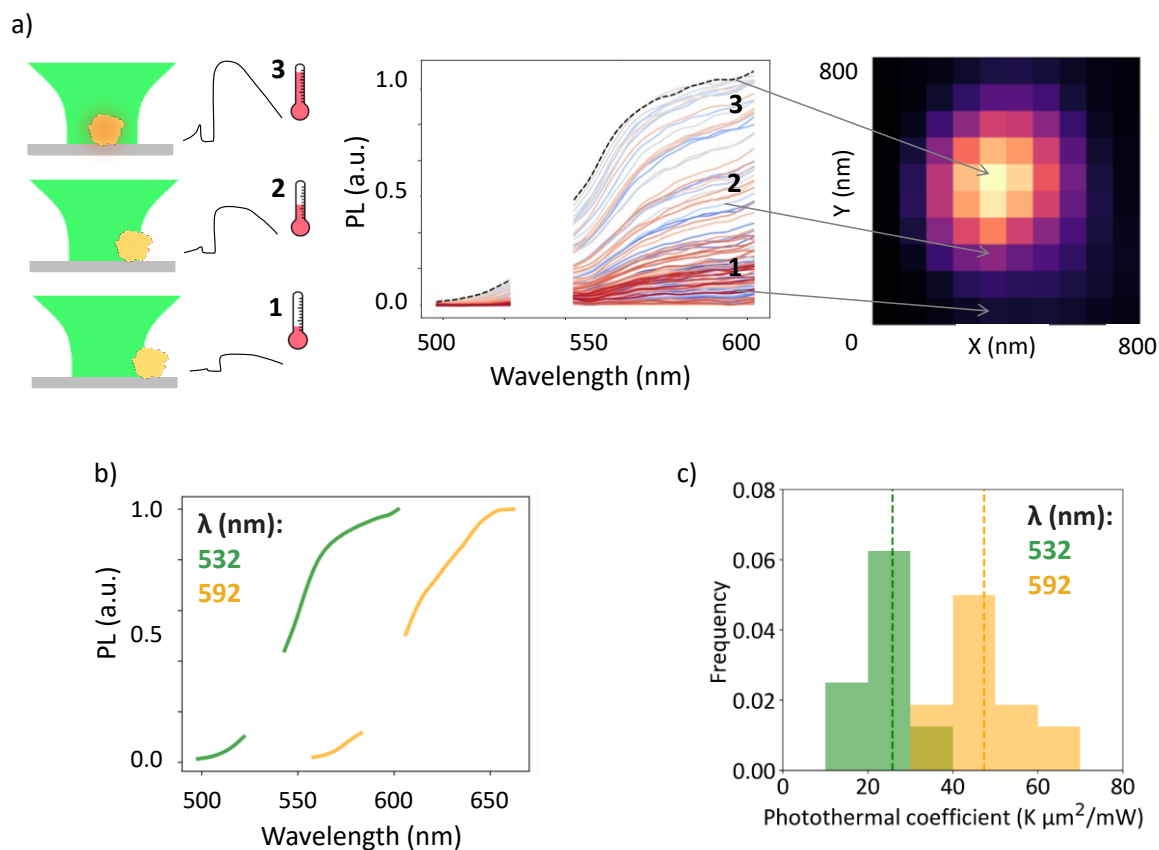


Figure S7. Measurement of AuNS temperature by confocal Hyperspectral anti-Stokes thermometry a) Scheme of the method employing a 532 nm laser, and the PL response from a AuNS pixel by pixel over a region of 800 nm x 800 nm. b) Normalized PL of a AuNS measured at two wavelengths: 532 and 592 nm. c) Histograms of the AuNS photothermal coefficient on water and supported onto glass, obtained at two wavelengths: 532 and 592 nm.

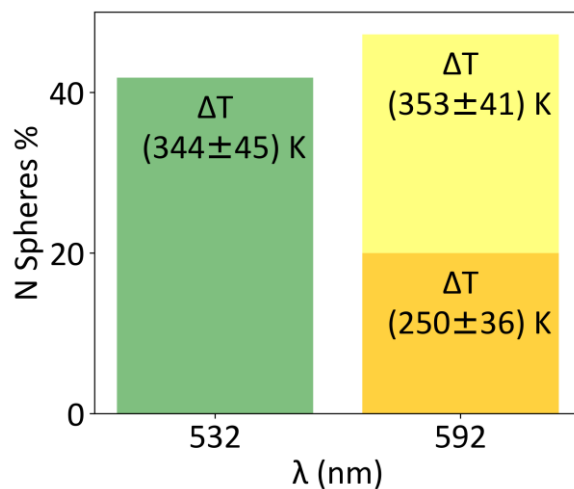


Figure S8. Percentage of spheres obtained as a function of the wavelength. The 592 nm laser was employed at two irradiances, the minimum to print (orange), and a higher one (yellow), such as the temperature increase was similar that the one obtained for the minimum printing irradiance necessary at 532 nm. The increment of temperature of each case was obtained through the measured photothermal coefficient.

Optical printing of AuNSs of different sizes

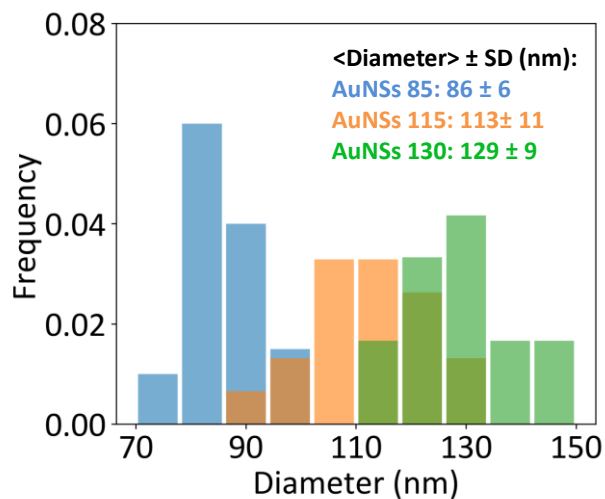


Figure S9. a) Histogram of the diameters of the three sized AuNS 85, 115 and 130 nm.

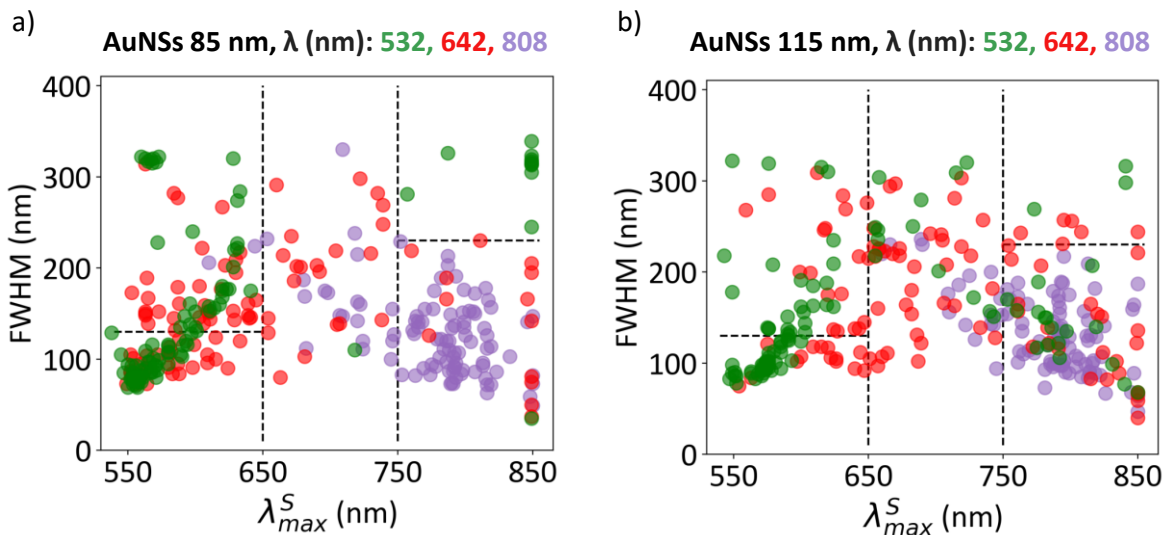


Figure S10. FWHM (nm) vs $\lambda_{\max}^S$ for the classification of 100 outcomes of the optical printing, the delimiter regions define the three groups: Spheres, Semistars/Aggregates or Stars as stated in Figure S4. Each dot color represents an outcome for printing depending on the laser wavelength. a) 100 outcomes for the optical printing at each laser of 85 nm AuNS. b) 100 outcomes for the optical printing at each laser of 115 nm AuNS.

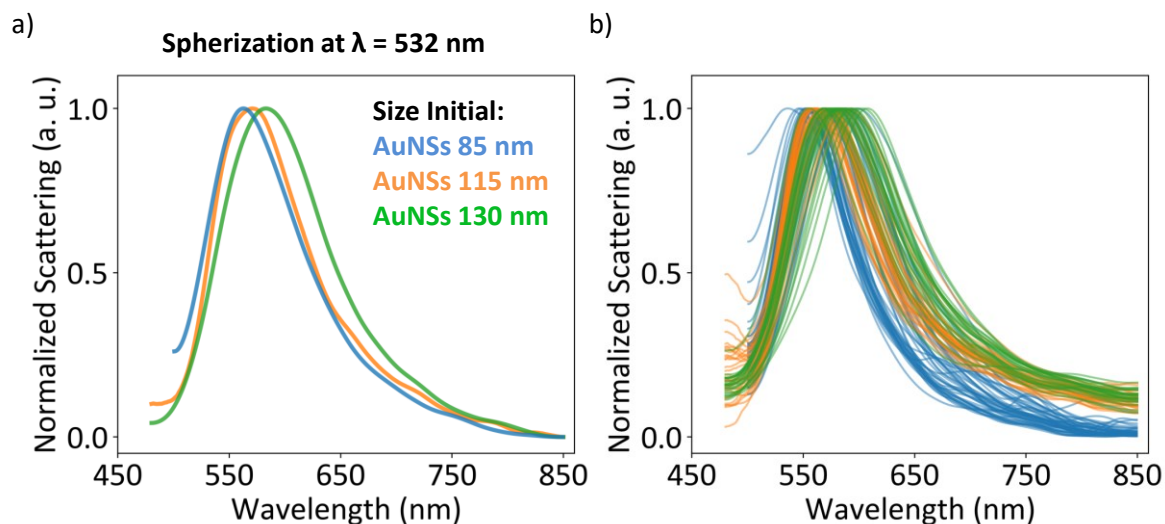


Figure S11. Reshaping to spheres for the three sized AuNS when printing at 532 nm. a) Average of scattering of total reshaped AuNS b) Scattering from the individual particles.